Biotechnology

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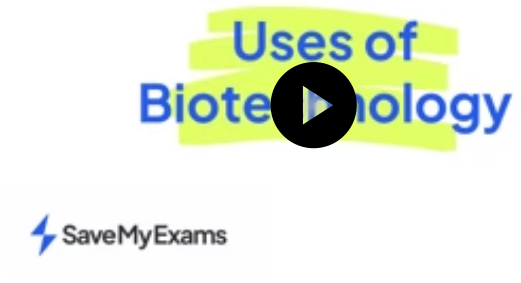

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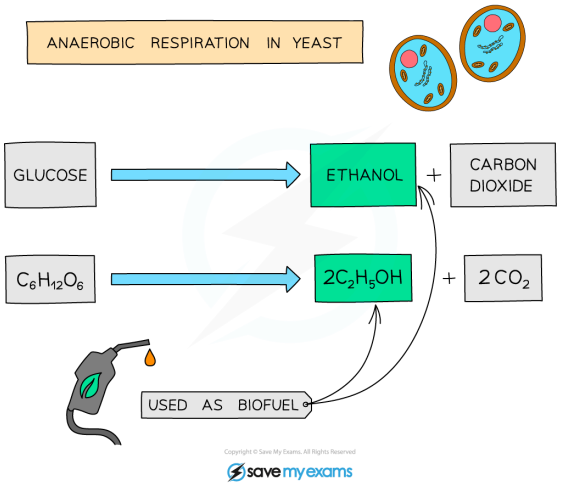
Everyday Products Made with Biotechnology

Biofuels

- **Yeast** is a single celled fungus that uses sugar as its food source
- Anaerobic respiration in yeast produces **ethanol** and **carbon dioxide**, as well as releasing energy for the yeast cell



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

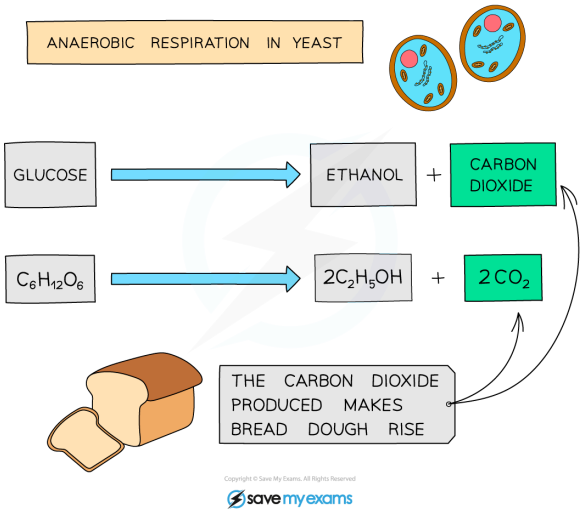
- The **ethanol** produced by anaerobic respiration in yeast can be used in the production of biofuels
  - In countries such as Brazil, biofuel is partly replacing petrol as the fuel for vehicles

- Biofuel production using yeast can be carried out as follows:
  1. **plant material** is broken up into small pieces and mixed with **yeast**
  2. yeast cells use the plant material as a substrate in anaerobic respiration and produce **ethanol** as a product
  3. the liquid is separated from the remaining solids and any water is removed, leaving a **concentrated solution of ethanol**
  4. the ethanol may then be blended with petrol, or used on its own as a fuel

- It is possible to use the **waste parts** of crop plants, such as the stalks or outer leaves, in biofuel production, but **crops may also be grown specifically** to be harvested for making ethanol
  - This is causing concern that there is **less land available** for local people to grow food crops needed for survival

Bread

- Yeast will **respire anaerobically** if it has access to plenty of **sugar**, even if oxygen is available
- This is taken advantage of in **bread making**, where the yeast is mixed with flour and water and respires anaerobically, producing **carbon dioxide**:



The carbon dioxide produced by fermentation (anaerobic respiration) of glucose is what makes bread dough rise

- The **carbon dioxide** produced by the yeast during respiration is caught in the dough, causing the bread to **rise**

Fruit Juice Production

lot more juice, but this does not break open all the cells so **a lot of juice is lost**
- By adding an enzyme called **pectinase** to the chopped up fruit, more juice is released
- Pectinase works by breaking down a chemical called **pectin** that is found inside plant cell walls
- Once pectin is broken down, the **cell walls break more easily** and more juice can be squeezed out of the fruit
- Adding pectinase to fruits also helps to **produce a clearer juice** as larger polysaccharides like pectin can make the juice seem cloudy - once they are broken down into smaller molecules, the juice becomes clearer

Biological Washing Powders

- Many stains on clothes are **organic molecules** – oil from skin, protein from blood, fat and protein from food
- Detergents that only contain soap can remove some of these stains when mixed with hot water, but it can take **a lot of time and effort and very high temperatures** to remove the stains entirely
- Biological washing powders contain **enzymes** similar to the digestive enzymes produced in the alimentary canal that help to break down large food molecules
- Using biological washing powders has several advantages, including:
  - **Quickly** breaking down **large, insoluble molecules** such as fats and proteins into **smaller, soluble ones** that **will dissolve** in washing water
  - They are **effective at lower temperatures**, meaning **less energy (and money)** has to be used in order to wash clothes to get them clean as washing water does not need to be heated to higher temperatures
  - They can be used to clean **delicate fabrics** that would not be suitable for washing at high temperatures

Other Uses of Biotechnology: Extended

- Lactose is the **sugar found in milk**
- Human babies are born with the ability to produce **lactase**, the **enzyme** that breaks down lactose
- In certain areas of the world, many people **lose the ability to produce lactase as they get older**
- This means that they can become **lactose intolerant** and react badly to the lactose in milk and products made from milk (cheese, yoghurt etc)
- Symptoms of lactose intolerance include **nausea, flatulence and diarrhoea** as their digestive system is upset by the lactose
- Milk can be made **lactose free** by **adding the enzyme lactase to it** and leaving it to stand for a while to allow the enzyme to break down the lactose

Lactose-free milk is a product made from adding the enzyme lactase to dairy milk to break down the sugars in it

Penicillin Production

- Penicillin was the **first antibiotic**, discovered in 1928 by **Alexander Fleming**
- He noticed that some bacteria he had left in a Petri dish had been killed by the naturally occurring ***Penicillium* mould**
- The penicillium mould produces a **chemical** to prevent it being infected by certain types of bacteria

Penicillin produced by the fungus Penicillium inhibits bacterial growth

- The chemical was isolated and named **penicillin**
- Since the discovery of penicillin, methods have been developed to **produce it on a large scale**, using an **industrial fermenter**

A diagram of an industrial fermenter used to produce large quantities of microorganisms

- **Fermenters** are containers used to grow ('**culture**') microorganisms like bacteria and fungi in **large amounts**
- These can then be used for many biotechnological processes like producing **genetically modified bacteria** and the **penicillium mould that produces penicillin**
- The advantage of using a fermenter is that **conditions can be carefully controlled** to produce large quantities of exactly the right type of microorganism

Condition	Why and how it is controlled
Aseptic precautions	Fermenter is cleaned by steam to kill microorganisms and prevent chemical contamination, which ensures only the desired microorganisms can grow
Nutrients	Nutrients are needed for use in respiration to release energy for growth and reproduction of the microorganisms
Optimum temperature	Temperature is monitored using probes and maintained using a water jacket. This ensures an optimum environment for enzymes to increase enzyme activity and prevent denaturation
Optimum pH	pH is monitored using a probe to check it is at the optimum value for the microorganism being grown. The pH can be adjusted using acids and alkalis
Oxygenation	Oxygen is required for aerobic respiration to take place
Agitation	Stirring paddles are used to ensure temperature, pH, nutrients, and oxygen are all distributed evenly throughout the fermenter

Mycoprotein – the process of creating food from a fungus

- The fungus Fusarium is cultured (grown) on an industrial scale in fermenters
- These fermenters are large vats that can be kept at the optimum pH and temperature for Fusarium to grow
- The fungus is grown in aerobic conditions (it is provided with oxygen) and provided with glucose syrup as a food source (to allow the fungus to respire)
- The fungus grows and multiplies within the fermenter
- The fungal biomass is then harvested and purified to produce mycoprotein
- Mycoprotein is a protein-rich food suitable for vegetarians eg. it is used in Quorn™ products

Production of Insulin

- Genetic modification is changing the genetic material of an organism by removing or altering genes within that organism, or by inserting genes from another organism
- The organism receiving the genetic material is said to be '**genetically modified**', or is described as a 'transgenic organism'
- The DNA of the organism that now contains DNA from another organism as well is known as 'recombinant DNA'
- The gene for human insulin has been inserted into bacteria which then produce **human insulin** which can be collected and purified for medical use to treat people with **diabetes**

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